AMRITA VISHWA VIDYAPEETHAM

AHEAD Online

Object Oriented Programming Using Java

Objectives

- Super and final Keyword.

Using Super

- Whenever a subclass needs to refer to its immediate superclass, it can do so by use of the keyword super.
- super has two general forms.
  - For calling the superclass' constructor.
  - To access a member of the superclass that has been hidden by a member of a subclass.

Calling Superclass Constructor

- A subclass can call a constructor defined by its superclass by use of super
  super(arg-list);

Example

```
class A{
int data1,data2;
A(int x,int y)
{
data1 = x;
data2 = y;
}}
```

Calling Superclass Constructor

```
class B extends A{
int data3;
B(int x,int y,int z)
{
super(x,y) // call superclass constructor
data3=z;
}}
```

Class B only needs to initialize the value unique to it, data3.

```
public class Main {
   public static void main(String[] args) {
      B obj= new B(1, 2, 3);
// Accessing the members of both superclass (A) and subclass (B)
      System.out.println("data1 from A: " + obj.data1);
      System.out.println("data2 from A: " + obj.data2);
      System.out.println("data3 from B: " + obj.data3);
   }
```

Calling Superclass Members

```
class A{
    void show(){
        System.out.println("Inside class A");
    }}
class B extends A{
    void show(){
        super.show();
        System.out.println("Inside class B");
    }}
```

Calling Superclass Members

- This usage has the following general form:

  super.*member*

- Member can be either a method or an instance variable.
- Useful when member names of a subclass hide members by the same name in the superclass

Final Keyword

- Used to restrict users
- It can be used with
    - Class
    - Method
    - variable

Final Keyword and Inheritance

```
class A {
final void meth() {
System.out.println("This is a final method.");
}
}
class B extends A {
void meth() { // ERROR! Can't override.
System.out.println("Illegal!");
}
}
```

- final keyword used to prevent method overriding.

Final Keyword and Inheritance

```
final class A {
// ... } // The following class is illegal.
class B extends A {
// ERROR! Can't subclass A // ...
 }
```

- A class declared as final can not be inherited.

Final Keyword and Inheritance

```
public class test {
    public static void main(String[] args) {
        final double PI = 3.14159;
        PI=PI*2;
        System.out.println("Value of PI: " + PI);
    }
}
```

Takeaways

- super keyword.
- final in inheritance.

Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition"*, *Oracle 2018*